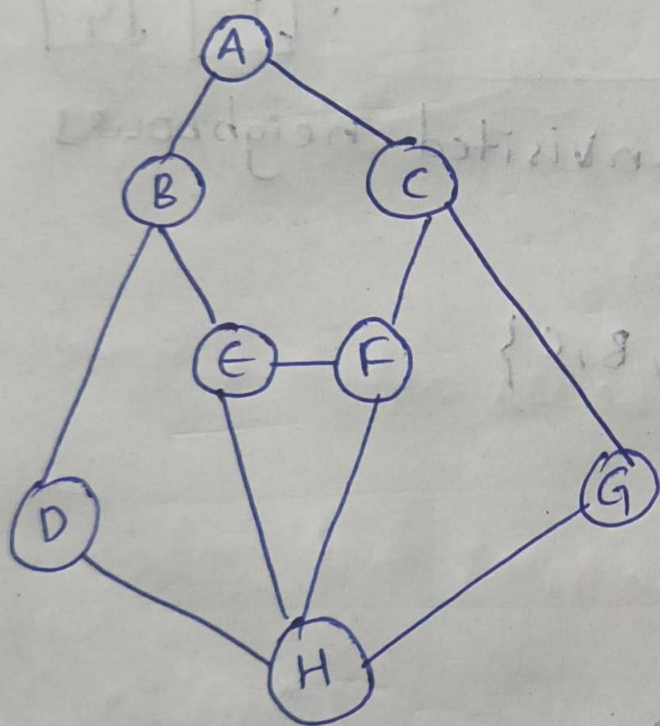


BFS:-

→ First the start vertex, then all of its neighbours, then their neighbours, queue

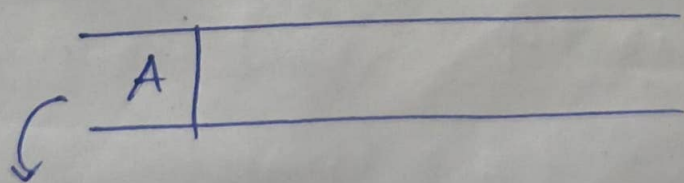
- 1. Enqueue first vertex
2. Dequeue vertex at front
3. If it's already visited, skip it
4. Otherwise mark it visited, output it and enqueue unvisited neighbours.
5. Repeat until queue is empty



A [B, C]
B [A, D, E]
C [A, F, G]
D [B, H]
E [B, F, H]
F [C, E, H]
G [C, H]
H [D, E, F, G]

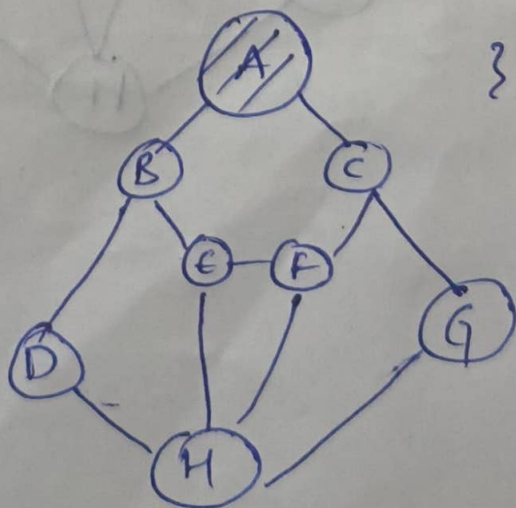
~~step~~

1.



mark A as
visited

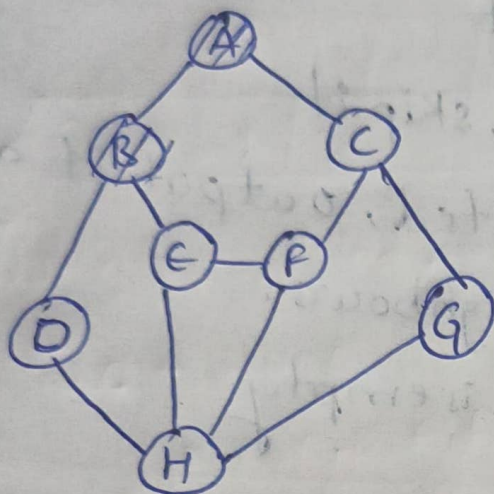
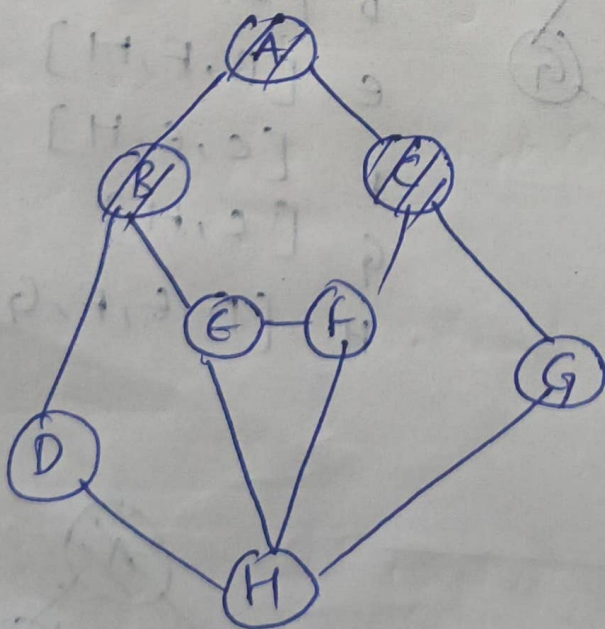
N: - B, C



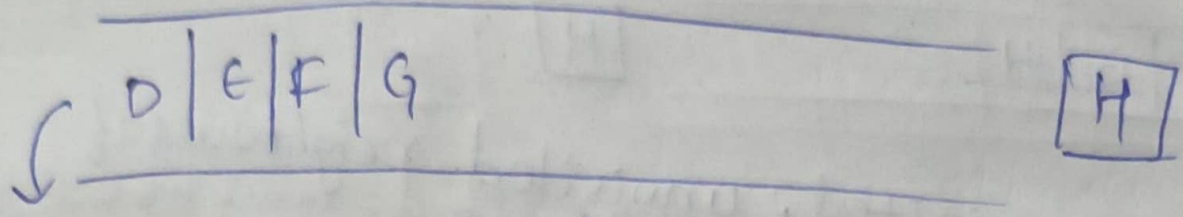
{A}

$B \quad C$

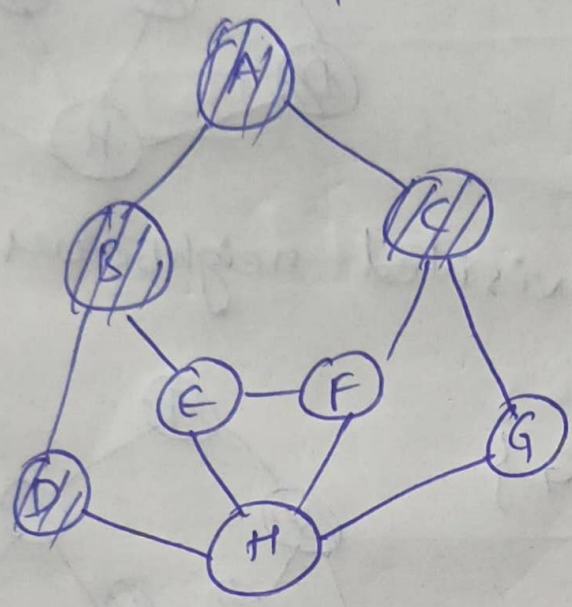
$D \quad E$

$$\{A, B\}$$

$$\{A, B, C\}$$


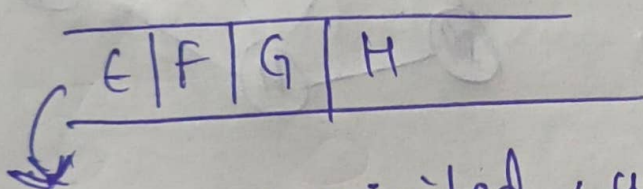
4.



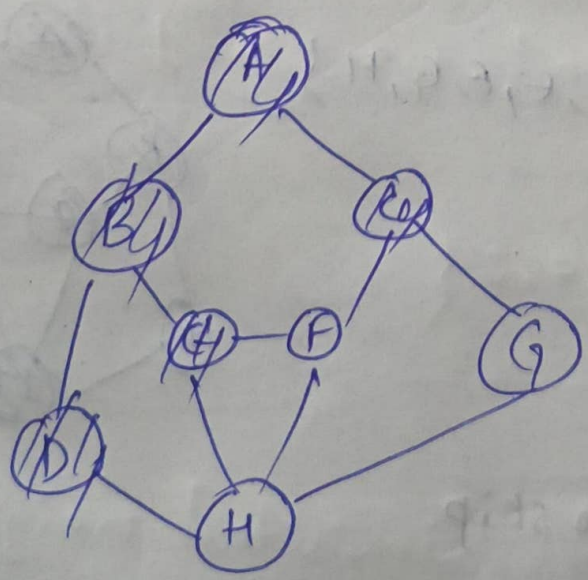
mark D as visited, unvisited neighbours of D are H. $\{A, B, C, D\}$



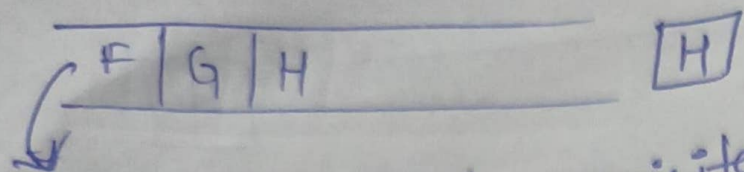
5.



mark E as visited, unvisited neighbours of E are already visited. $\{A, B, C, D, E\}$



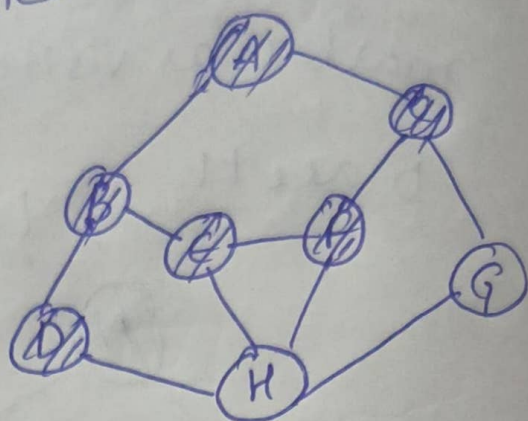
6.



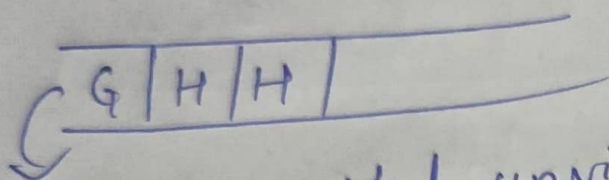
mark F as visited, unvisited neighbours of

F are H

$\{A, B, C, D, E, F\}$



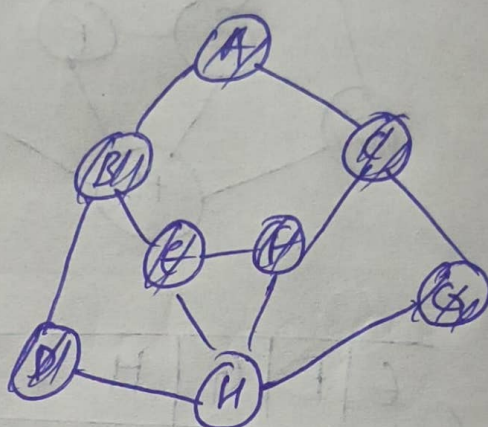
7.



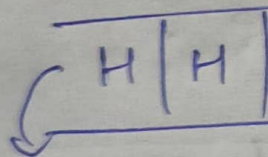
mark G as visited, unvisited neighbours of

G are none

$\{A, B, C, D, E, F, G\}$



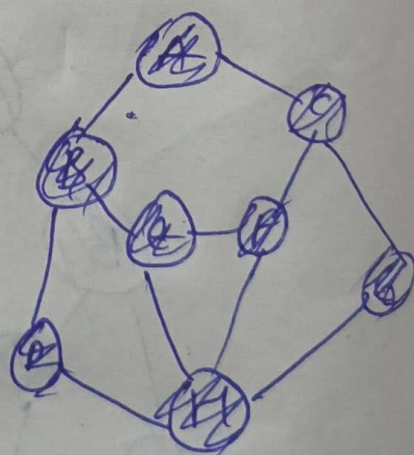
8.



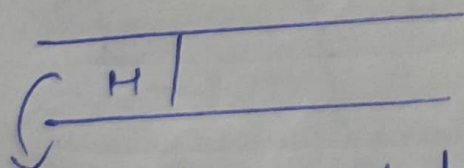
no unvisited neighbours ~~at~~, mark H

as visited.

$\{A, B, C, D, E, F, G, H\}$



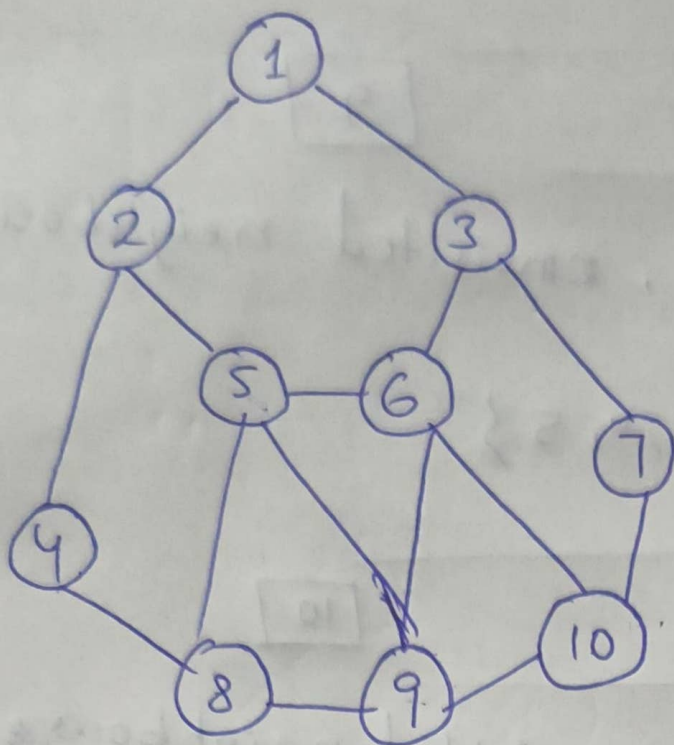
9.



already visited so skip.

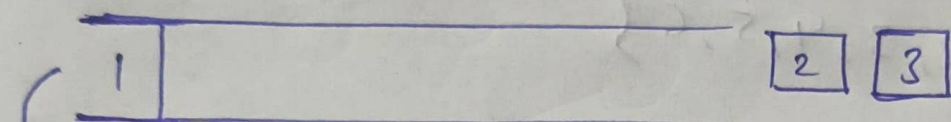
adjacency list :- $\{A, B, C, D, E, F, G, H\}$

2.



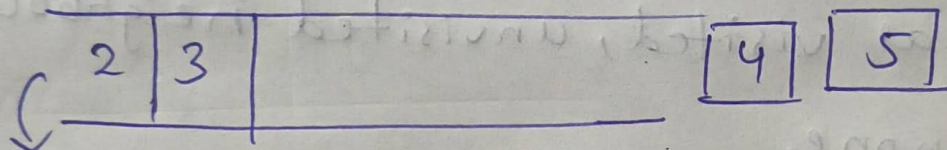
- 1 [2, 3]
- 2 [1, 4, 5]
- 3 [1, 6, 7]
- 4 [2, 8]
- 5 [2, 6, 8, 9]
- 6 [3, 5, 9, 10]
- 7 [3, 10]
- 8 [4, 5, 9]
- 9 [5, 6, 8, 10]
- 10 [6, 7, 9]

1.



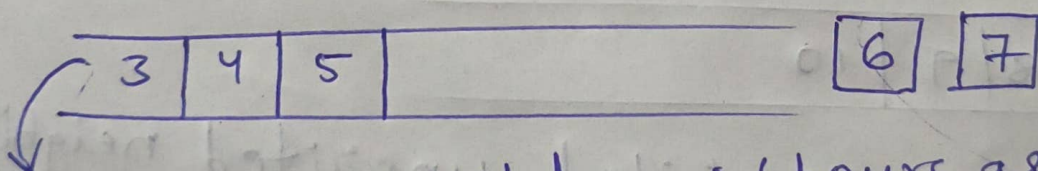
mark 1 as visited, 2, 3 as unvisited. {1}

2.



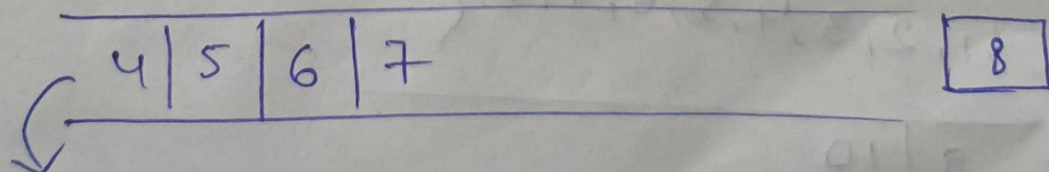
mark 2 as visited, neighbours are already visited, {1, 2}

3.



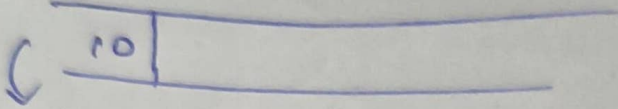
mark 3 as visited, neighbours are already visited. {1, 2, 3}

4.



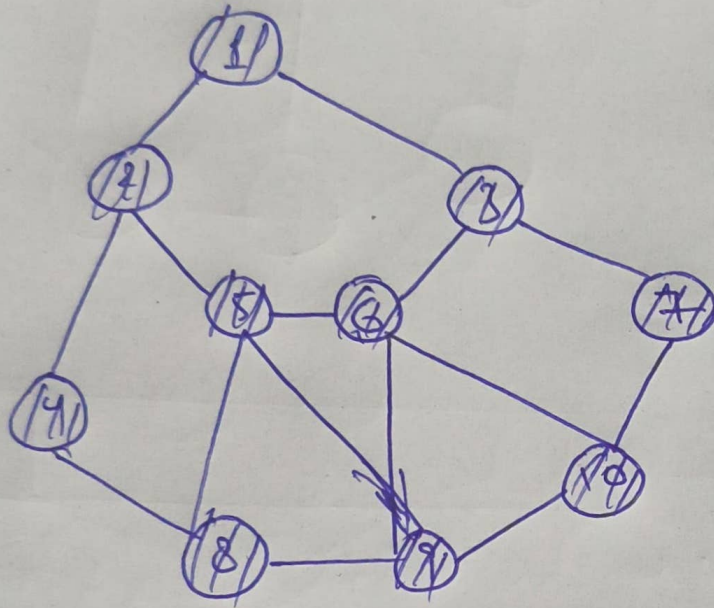
{1, 2, 3, 4}
mark 4 as visited, neighbours are ~~already~~ 8 which is not visited

10.



mark 10 as visited, unvisited neighbours are none

{ 1, 2, 3, ~~4~~, 4, 5, 6, 7, 8, 9, 10 }



adjacency list: - { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 }